

(3) Let x be a random variable with p.d.f.

$$f(x) = \begin{cases} ax & ; 0 \leq x \leq 1 \\ a & ; 1 \leq x \leq 2 \\ -ax + 3a & ; 2 \leq x \leq 3 \\ 0 & ; \text{otherwise} \end{cases}$$

Find:

i) the value of a .

ii) $P(x \leq 1.5)$

iii) $E(x)$

iv) $V(x)$

v) $E(3x+5)$, $V(3x+1)$.

sol Here

By the $f(x) = \begin{cases} ax & ; 0 \leq x \leq 1 \\ a & ; 1 \leq x \leq 2 \\ -ax + 3a & ; 2 \leq x \leq 3 \\ 0 & ; \text{otherwise} \end{cases}$

i) By the definition of pdf,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^1 ax dx + \int_1^2 a dx + \int_2^3 (-ax + 3a) dx = 1$$

$$\Rightarrow \left[\frac{ax^2}{2} \right]_0^1 + \left[ax \right]_1^2 + \left[-\frac{ax^2}{2} + 3ax \right]_2^3 = 1$$

$$\Rightarrow \frac{a}{2} + 2a - a + \frac{-9a}{2} + \frac{18a}{2} + 9a - 6a = 1$$

$$\Rightarrow \frac{a - 9a + 4a}{2} + 2a - a + 9a - 6a = 1$$

$$\Rightarrow \frac{-4a}{2} + 4a = 1$$

$$\Rightarrow 2a = 1$$

$$\therefore a = \frac{1}{2}$$

Hence, the value of a is $\frac{1}{2}$. Ans

$$\therefore f(x) = \begin{cases} \frac{1}{2}x & ; 0 \leq x \leq 1 \\ \frac{1}{2} & ; 1 \leq x \leq 2 \\ -\frac{1}{2}x + \frac{3}{2} & ; 2 \leq x \leq 3 \\ 0 & ; \text{otherwise} \end{cases}$$

$$\begin{aligned} \text{ii)} P(x \leq 1.5) &= \int_0^{1.5} f(x) \cdot dx = \int_0^1 \frac{1}{2}x \, dx + \int_1^{1.5} \frac{1}{2} \, dx \\ &= 0.5 \quad \underline{\text{Ans}} \end{aligned}$$

$$\begin{aligned} \text{iii)} E(x) &= \int_{-\infty}^{\infty} x f(x) \, dx = \int_0^1 \frac{1}{2}x^2 \, dx + \int_1^2 \frac{1}{2}x \, dx + \\ &\quad \int_2^3 \left(-\frac{1}{2}x^2 + \frac{3}{2}x\right) \, dx = 1.5 \quad \underline{\text{Ans}} \end{aligned}$$

⑩

$$\text{iii)} \quad V(X) = E(X^2) - [E(X)]^2$$

$$= \int_0^1 x^2 f(x) dx + \int_1^2 x^2 f(x)$$

$$= \int_0^1 x^2 \cdot \frac{1}{2} x dx + \int_1^2 x^2 \cdot \frac{1}{2} dx + \int_2^3 x^2 \left(\frac{1}{2} x + \frac{3}{2} \right) dx - (1.5)^2$$

$$= \int_0^1 \frac{1}{2} x^3 dx + \int_1^2 \frac{x^2}{2} dx + \int_2^3 \left(\frac{1}{2} x^3 + \frac{3}{2} x^2 \right) dx - (1.5)^2$$

$$= 2.67 - (1.5)^2$$

$$\therefore V(X) = 0.416 \quad \underline{\text{Ans}}$$

$$\text{iv)} \quad E(3X+5) = 3E(X) + E(5)$$

$$= 3 \times 1.5 + 5$$

$$= 9.5 \quad \underline{\text{Ans}}$$

$$V(3X+5) = 3^2 V(X) + V(5)$$

$$= 9 \times 0.416 + 0$$

$$= 3.744 \quad \underline{\text{Ans}}$$

$$(\because V(c) = 0)$$

2020-Fall / 2021-Spring

④

3.a) Let X be a continuous random variable with p.d.f.

$$p(x) = \begin{cases} ax & ; 0 \leq x \leq 2 \\ a & ; 2 \leq x \leq 4 \\ ax+3a & ; 4 \leq x \leq 6 \\ 0 & ; \text{otherwise} \end{cases}$$

i) Determine constant a .

ii) Find $P(X \leq 1.5)$ and $P(1.5 \leq X \leq 2.5)$

iii) Obtain $E(X)$.

~~sof~~ Here,

$$f(x) = \begin{cases} ax & ; 0 \leq x \leq 2 \\ a & ; 2 \leq x \leq 4 \\ ax+3a & ; 4 \leq x \leq 6 \\ 0 & ; \text{otherwise} \end{cases}$$

i) By the definition of p.d.f, we get:

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^2 ax dx + \int_2^4 a dx + \int_4^6 (ax+3a) dx = 1$$

$$\Rightarrow \left[\frac{ax^2}{2} \right]_0^2 + \left[ax \right]_2^4 + \left[\frac{ax^2}{2} + 3ax \right]_4^6 = 1$$

$$\Rightarrow a \cdot \frac{2^2}{2} + 4a - 2a + \frac{36a}{2} + 18a - \frac{16a}{2} - 12a = 1$$

$$\Rightarrow \frac{4a}{2} + \frac{36a}{2} - \frac{16a}{2} + 4a - 2a + 18a - 12a = 1$$

$$\Rightarrow 12a + 8a = 1$$

$$\Rightarrow 20a = 1$$

$$\therefore a = \frac{1}{20}$$

The value of constant a is $\frac{1}{20}$ Ans

Hence,

$$f(x) = \begin{cases} \frac{1}{20} & 0 \leq x \leq 2 \\ \frac{1}{20} & 2 \leq x \leq 4 \\ \frac{4}{20} + \frac{3}{20}x & 4 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{ii) } P(X \leq 1.5) = \int_0^{1.5} \frac{x}{20} dx = \frac{1}{20} \left[\frac{x^2}{2} \right]_0^{1.5} = 0.05625 \text{ Ans}$$

$$P(1.5 \leq X \leq 2.5) = \int_{1.5}^2 \frac{x}{20} dx + \int_2^{2.5} \frac{1}{20} dx = 0.125 \text{ Ans}$$

~~= 0.4~~

$$\text{iii) } E(X) = \int_{-\infty}^{\infty} x f(x) dx = \int_0^2 \frac{x^2}{20} dx + \int_2^4 \frac{x}{20} dx + \int_4^6 \left(\frac{x^2}{20} + \frac{3x}{20} \right) dx$$

$$= 4.467 \text{ Ans}$$

apter: 4 Random Variable and Probability distribution

2018-Spring

Python, NAST

c) A continuous random variable X has the following density function.

$$f(x) = \begin{cases} Kx^2 & \text{for } 0 < x < 1 \\ 0 & \text{; otherwise} \end{cases}$$

i) Find the value of K .

ii) $P(0 < X < 0.7)$

iii) $E(X)$ and $V(X)$.

Solⁿ Here,

$$f(x) = \begin{cases} Kx^2 & ; 0 < x < 1 \\ = 0 & ; \text{otherwise} \end{cases}$$

i) By the definition of pdf,

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^1 Kx^2 dx = 1$$

$$\Rightarrow K \left[\frac{x^3}{3} \right]_0^1 = 1$$

$$\Rightarrow K \cdot \frac{1}{3} = 1$$

$$\therefore K = 3$$

Hence the value of K is 3. Ans

$$\therefore f(x) = \begin{cases} 3x^2 & ; 0 < x < 1 \\ = 0 & ; \text{otherwise} \end{cases}$$

$$\begin{aligned}
 \text{ii) } P(0 \leq x \leq 0.7) &= \int_0^{0.7} f(x) dx \\
 &= \int_0^{0.7} 3x^2 dx \\
 &= 0.343 \text{ Ans}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } E(x) &= \int_x x f(x) dx = \int_0^1 x \cdot 3x^2 dx \\
 &= \int_0^1 3x^3 dx = 0.75
 \end{aligned}$$

Also,

$$V(x) = E(x^2) - [E(x)]^2$$

$$= \int_x x^2 f(x) dx - [E(x)]^2$$

$$= \int_0^1 x^2 \cdot 3x^2 dx - (0.75)^2$$

$$= \int_0^1 3x^4 dx - (0.75)^2$$

$$\therefore V(x) = 0.0375 \text{ Ans}$$

2016-Fall

3. a) Suppose that x is a random variable, whose probability density function is given by

$$f(x) = c(4x - 2x^2) \quad ; \quad 0 < x < 2 \\ = 0 \quad ; \quad \text{otherwise}$$

i) Determine the value of c .

ii) Determine the cumulative distribution function of x .

iii) Find $P(x > 1)$.

~~sol~~ Here,

$$f(x) = c(4x - 2x^2) \quad ; \quad 0 < x < 2 \\ = 0 \quad ; \quad \text{otherwise}$$

i) By the definition of p.d.f.,

$$\int_x f(x) dx = 1$$

$$\Rightarrow \int_0^2 c(4x - 2x^2) dx = 1$$

$$\Rightarrow c \int_0^2 (4x - 2x^2) dx = 1$$

$$\Rightarrow c \times 2.6667 = 1$$

$$\Rightarrow \therefore c = \frac{1}{2.6667} = \frac{3}{8}$$

$$\therefore c = \frac{3}{8}$$

Hence, the required value of c is $\frac{3}{8}$. Ans

$$\therefore f(x) = \frac{3}{8}(4x - 2x^2) \quad ; \quad 0 < x < 2 \\ = 0 \quad ; \quad \text{otherwise}$$

ii) The cumulative distribution of x is given by

$$F(x) = \int_0^x f(x) dx = \int_0^x \frac{3}{8} (4x - 2x^2) dx$$

$$= \frac{3}{8} \left[\frac{4x^2}{2} - \frac{2x^3}{3} \right]_0^x$$

$$= \frac{3}{8} \left(\frac{4x^2}{2} - \frac{2x^3}{3} - \frac{4 \times 1500^2}{2} + \frac{2 \times 1500^3}{3} \right)$$

$$\therefore f(x) = \frac{3}{8} \left(\frac{4x^2}{2} - \frac{2x^3}{3} \right) \quad \text{Ans}$$

$$\text{iii) } P(x > 1): \int_1^2 \frac{3}{8} (4x - 2x^2) dx = 0.5 \quad \text{Ans}$$

2009-fall

20) A random variable x has the density function $f(x) = a + bx^2$, $0 < x < 1$. Determine a and b show that, mean is $\frac{2}{3}$. Also find the variance of the distribution.

Here,

$$f(x) = a + bx^2 \quad ; \quad 0 < x < 1.$$

$$\text{mean, } E(x) = \frac{2}{3}$$

Now, By the definition of p.d.f,

$$\Rightarrow \int_x f(x) dx = \int_0^1 (a + bx^2) dx = 1$$

$$\Rightarrow \left[\frac{ax}{1} + \frac{bx^3}{3} \right]_0^1 = 1$$

$$\Rightarrow a + \frac{b}{3} = 1$$

$$\therefore 3a + b = 3 \quad \text{--- (1)}$$

Also,

$$E(x) = \int_x x f(x) dx = \int_0^1 x(a + bx^2) dx = \int_0^1 (ax + bx^3) dx$$

$$\Rightarrow \frac{2}{3} = \left[\frac{ax^2}{2} + \frac{bx^4}{4} \right]_0^1 = \frac{a}{2} + \frac{b}{4} = \frac{4a + 2b}{8}$$

$$\therefore 12a + 6b = 16 \quad \text{--- (2)}$$

on solving (1) & (2), we get:-

$$a = \frac{1}{3} ; b = 2$$

Hence, the required value of a and b is:

$$a = \frac{1}{3} ; b = 2 \quad \underline{\text{Ans}}$$

Also,

$$\therefore f(x) = \frac{1}{3} + 2x^2; \quad 0 < x < 1$$

Now,

$$\text{Variance of } x = V(x) = E(x^2) - [E(x)]^2$$

$$= \int_0^1 x^2 f(x) dx - [E(x)]^2$$

$$= \int_0^1 x^2 \left(\frac{1}{3} + 2x^2 \right) dx - \left(\frac{2}{3} \right)^2$$

$$= \int_0^1 \left(\frac{1}{3} x^2 + 2x^4 \right) dx - \left(\frac{2}{3} \right)^2$$

$$\therefore V(x) = \cancel{0.067} = \frac{2}{30} = \frac{1}{15} \text{ Ans}$$

2017 - Fall

- (16) The length of time (in minute) that is a certain lady speaks on the telephone is found to be a random phenomenon with the probability function specified by the probability density function $f(x) = \begin{cases} Ae^{-x/5} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$
- i) Find the value of A.
- ii) What is the probability that the number of minute that she will talk over the phone is
- a. more than 10 minute
 - b. less than 5 minute
 - c. between 5 and 10 minute
- iii) Find mean and variance of talk time on the telephone.

Sol Here,

$$f(x) = \begin{cases} Ae^{-x/5} & \text{for } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

i) By the definition of p.d.f, we get:

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^{\infty} Ae^{-x/5} dx = 1$$

$$\Rightarrow A \int_0^{\infty} e^{-x/5} dx = 1$$

$$\Rightarrow A \cdot \left[\frac{e^{-x/5}}{-1/5} \right]_0^{\infty} = 1$$

$$\Rightarrow -5A [e^{-\infty} - e^{-0}] = 1$$

$$\Rightarrow -5A (0 - 1) = 1$$

$$\Rightarrow 5A = 1$$

$$\therefore A = 1/5$$

Hence, the value of A is $1/5$.

ii) $\therefore f(x) = \frac{1}{5} e^{-x/5}$ for $x \geq 0$

a. The probability that the number of minutes that she will talk over the phone more than 10 minutes B1

$$\begin{aligned} P(X > 10) &= \int_{10}^{\infty} f(x) dx = \int_{10}^{\infty} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_{10}^{\infty} = \frac{1}{5} \times -5 (e^{-\infty} - e^{-10/5}) \\ &= \cancel{(-e^{-\infty})} - (0 - e^{-10/5}) \\ &= e^{-2} \quad \underline{\text{Ans}} \end{aligned}$$

b. The probability that the number of minutes that she will talk over the phone ~~more than~~ less than 5 minutes: $P(X < 5) = \int_0^5 f(x) dx = \int_0^5 \frac{1}{5} e^{-x/5} dx$

$$\begin{aligned} &= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_0^5 = \frac{-5}{5} (e^{-5/5} - e^{-0}) \\ &= -(e^{-1} - 1) \\ &= 1 - e^{-1} \quad \underline{\text{Ans}} \end{aligned}$$

c. The probability that the number of minutes that she will talk over the phone between 5 and 10 minutes B1

$$\begin{aligned} P(5 < X < 10) &= \int_5^{10} f(x) dx = \int_5^{10} \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \left[\frac{e^{-x/5}}{-1/5} \right]_5^{10} = - (e^{-10/5} - e^{-5/5}) \\ &= (e^{-1} - e^{-2}) \quad \underline{\text{Ans}} \end{aligned}$$

iii >

$$\text{Mean} = E(X) = \int_0^{\infty} \frac{1}{5} x f(x) dx = \int_0^{\infty} x \cdot \frac{1}{5} e^{-x/5} dx$$

$$= \frac{1}{5} \int_0^{\infty} x e^{-x/5} dx$$

~~put $x = t$~~

$$= \frac{1}{5} \left[\frac{x \cdot e^{-x/5}}{-1/5} - \left(\int_0^{\infty} 1 \cdot e^{-x/5} dx \right) \right]_0^{\infty}$$

~~$\therefore \frac{dx}{dt} = 1$~~

$$= \frac{1}{5} \left[-5x e^{-x/5} - \int_0^{\infty} 1 \cdot (-5 e^{-x/5}) dx \right]_0^{\infty}$$

$$= \frac{1}{5} \left[-5x e^{-x/5} + 5 \frac{e^{-x/5}}{-1/5} \right]_0^{\infty}$$

$$= \frac{1}{5} \left[-5x e^{-x/5} - 25 e^{-x/5} \right]_0^{\infty}$$

$$= \frac{1}{5} \left[-5 \times \infty \times e^{-\infty} + 0 - 25 e^{-\infty} + 25 e^0 \right]$$

$$= \frac{1}{5} (0 + 0 - 0 + 25)$$

$$= \frac{25}{5}$$

$$\therefore E(X) = 5 \quad \underline{\text{Ans}}$$

$$\begin{aligned}\therefore E(x^2) &= \int_0^{\infty} x^2 f(x) dx = \int_0^{\infty} x^2 \frac{1}{5} e^{-x/5} dx \\ &= \frac{1}{5} \int_0^{\infty} x^2 e^{-x/5} dx \\ &= \frac{1}{5} \left[x^2 \int_0^{\infty} e^{-x/5} dx - \int_0^{\infty} \frac{d}{dx} x^2 \int_0^{\infty} e^{-x/5} dx \right]\end{aligned}$$

$$= \frac{1}{5} \left[x^2 \int_0^{\infty} e^{-x/5} dx - \int_0^{\infty} 2x \cdot \frac{e^{-x/5}}{-1/5} dx \right]$$

$$= \frac{1}{5} \left[x^2 \int_0^{\infty} \frac{e^{-x/5}}{5} dx + 10 \int_0^{\infty} \frac{x e^{-x/5}}{5} dx \right]$$

$$= \left[\frac{x^2 e^{-x/5}}{-1/5} + 10 \times \frac{1}{5} \right]_0^{\infty}$$

$$= 0 - (\infty e^{-\infty} - 0e^0 + 50)$$

$$\therefore E(x^2) = 50$$

$$\begin{aligned}\therefore V(x) &= E(x^2) - [E(x)]^2 \\ &= 50 - 25^2 \\ &= 25 \quad \underline{\text{Ans}}\end{aligned}$$

2009 spring

- ① Two balanced dice are thrown, find the probability distribution and the expected value of the sum of numbers of point on them. Also find the variance.

Let x be a random variable that denotes the sum of numbers on two dice.

Sample points	Sum of number	Probability
(1, 1)	2	$\frac{1}{36}$
(1, 2), (2, 1)	3	$\frac{2}{36}$
(1, 3), (2, 2), (3, 1)	4	$\frac{3}{36}$
(2, 3), (1, 4), (4, 1), (3, 2)	5	$\frac{4}{36}$
(3, 3), (1, 5), (2, 4), (4, 2), (5, 1)	6	$\frac{5}{36}$
(1, 6), (3, 4), (4, 3), (5, 2), (2, 5), (6, 1)	7	$\frac{6}{36}$
(2, 6), (4, 4), (5, 3), (3, 5), (6, 2)	8	$\frac{5}{36}$
(3, 6), (6, 3), (5, 4), (4, 5)	9	$\frac{4}{36}$
(4, 6), (6, 4), (5, 5)	10	$\frac{3}{36}$
(6, 5), (5, 6)	11	$\frac{2}{36}$
(6, 6)	12	$\frac{1}{36}$

~~Here~~ Total probability = 1

$$\begin{aligned}\text{The expected value, } E(x) &= 2 \times \frac{1}{36} + 3 \times \frac{2}{36} + 4 \times \frac{3}{36} + 5 \times \frac{4}{36} + 6 \times \frac{5}{36} \\ &\quad + 7 \times \frac{6}{36} + 8 \times \frac{5}{36} + 9 \times \frac{4}{36} + 10 \times \frac{3}{36} + 11 \times \frac{2}{36} \\ &\quad + 12 \times \frac{1}{36} = 7 \quad \underline{\underline{\text{Ans}}}\end{aligned}$$

Now,

$$V(X) = E(X^2) - [E(X)]^2$$

where,

$$\begin{aligned} E(X^2) &= 2^2 \times \frac{1}{36} + 3^2 \times \frac{2}{36} + 4^2 \times \frac{3}{36} + 5^2 \times \frac{4}{36} + 6^2 \times \frac{5}{36} + \\ &\quad 7^2 \times \frac{6}{36} + 8^2 \times \frac{8}{36} + 9^2 \times \frac{4}{36} + 10^2 \times \frac{3}{36} + 11^2 \times \frac{2}{36} \\ &\quad + 12^2 \times \frac{1}{36} = \frac{1974}{36} = \frac{329}{6} \end{aligned}$$

$$\therefore V(X) = \frac{329}{6} - 7^2 = \frac{35}{6} = 5.83 \text{ Ans}$$

2017-Fall / 2013-Spring / 2023-Spring,

~~2017-Spring / 20~~

3. a) The unbiased coin is tossed until a head appears. What is the expectation of number of tosses required to get the first head?

Solⁿ Let x be the random variable that denotes the number of tosses required to get the first head.

Expectation of ~~tosses~~ x , $E(x) = \sum_{i=1}^{\infty} x_i P(x_i)$

Now,

Outcomes	No. of tosses (x_i)	$P(x_i)$
H	1	$\frac{1}{2}$
TH	2	$\frac{1}{4}$
TTH	3	$\frac{1}{8}$
TTTH	4	$\frac{1}{16}$
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$

Now,

The expected value of x is $= \sum_{i=1}^{\infty} x_i P(x_i)$

$$= 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + \dots$$

$$\therefore E(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \quad (1)$$

Let $S = E(x)$

$$\therefore S = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \quad (2)$$

$$\therefore \frac{S}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{64} + \dots \quad (3)$$

From (2) and (3), we get:-

$$S - \frac{S}{2} = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} - \frac{1}{4} - \frac{2}{8} - \frac{3}{16} - \frac{4}{64} - \dots$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4} \right) + \left(\frac{3}{8} - \frac{2}{8} \right) + \left(\frac{4}{16} - \frac{3}{16} \right) + \dots$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\Rightarrow \frac{1}{2} S = \frac{1/2}{1 - 1/2} \quad \left(\because S_{\infty} = \frac{at_1}{1-r} \right)$$

$$\Rightarrow \frac{1}{2} S = \frac{1}{2} \times 2$$

$$\therefore S = 2$$

Hence, the required expectation of number of tosses is 2. Ans

2016 - Spring

2. From a lot of 10 items containing 3 defectives a sample of 4 item is drawn at random. Let the random variable X denote the number of defective items in the sample. Answer the following:

i) $E(X)$

ii) $V(X)$

iii) The probability distribution of X .

* Let X be a random variable that denotes the number of defective item in the sample.

$$X = 0, 1, 2, 3.$$

Probability of getting 0 defective item:

$$P(X=0) = \frac{{}^3C_0 \cdot {}^7C_4}{{}^{10}C_4} = \frac{1}{6}$$

Probability of getting 1 defective item:

$$P(X=1) = \frac{{}^3C_1 \cdot {}^7C_3}{{}^{10}C_4} = \frac{1}{2}$$

Probability of getting 2 defective item:

$$P(X=2) = \frac{{}^3C_2 \cdot {}^7C_2}{{}^{10}C_4} = \frac{3}{10}$$

Probability of getting 3 defective item:

$$P(X=3) = \frac{{}^3C_3 \cdot {}^7C_1}{{}^{10}C_4} = \frac{1}{30}$$

$$i) E(X) = 0 \times \frac{1}{16} + 1 \times \frac{1}{2} + 2 \times \frac{3}{10} + 3 \times \frac{1}{30} = \frac{6}{5} \text{ Ans}$$

$$ii) V(X) = E(X^2) - [E(X)]^2$$

$$= 0^2 \times \frac{1}{16} + 1^2 \times \frac{1}{2} + 2^2 \times \frac{3}{10} + 3^2 \times \frac{1}{30} - \left(\frac{6}{5}\right)^2$$

$$= 0.56 \text{ Ans}$$

iii) The probability distribution of X is given by

$P(X=x_i)$	0	1	2	3	Total
$P(x)$	$\frac{1}{16}$	$\frac{1}{2}$	$\frac{3}{10}$	$\frac{1}{30}$	1

③ Urn A contains 5 cards numbered from 1 to 5 and urn B contains 4 cards numbered from 1 to 4. One card is drawn from each of two urns. Find the probability function of sum of the number which appears on the cards drawn and its mathematical expectation.

29. Let X be a random variable that denotes the number sum of the number on the urns.

If one card is drawn from each of two urns. The probability function is given as below:

outcomes	sum of numbers	probability
(1, 1)	2	$\frac{1}{20}$
(1, 2), (2, 1)	3	$\frac{2}{20}$
(1, 3), (2, 2), (3, 1)	4	$\frac{3}{20}$
(1, 4), (2, 3), (3, 2), (4, 1)	5	$\frac{4}{20}$
(2, 4), (3, 3), (4, 2), (5, 1), (1, 5)	6	$\frac{4}{20}$
(3, 4), (4, 3), (5, 2), (2, 5)	7	$\frac{3}{20}$
(5, 3), (4, 4), (3, 5)	8	$\frac{2}{20}$
(4, 5) , (5, 4) (5, 4)	9	$\frac{1}{20}$
6		

$$E(X) = 2 \times \frac{1}{20} + 3 \times \frac{2}{20} + 4 \times \frac{3}{20} + 5 \times \frac{4}{20} + 6 \times \frac{4}{20} + 7 \times \frac{3}{20} + 8 \times \frac{2}{20} + 9 \times \frac{1}{20} = \frac{11}{2} \text{ Ans}$$

Discrete R.V

① A random variable X has following probability distribution

Value of $X: x$	-2	-1	0	1	2	3
$P(x)$	0.1	K	0.2	$2K$	0.3	K

Find:

i) the value of K .

ii) $P(X = -1)$

iii) $P(X \leq 2)$

iv) $P(-1 < X < 3)$

v) The expected value of random variable.

sol Here,

value of $X: x$	-2	-1	0	1	2	3
$P(x)$	0.1	K	0.2	$2K$	0.3	K

i) ~~since~~ By the definition of pmf,

$$\sum_x P(x) = 1$$

$$\Rightarrow 0.1 + K + 0.2 + 2K + 0.3 + K = 1$$

$$\Rightarrow 4K + 0.6 = 1$$

$$\Rightarrow 4K = 1 - 0.6 = 0.4$$

$$\therefore K = 0.1$$

Hence, the value of K is 0.1 Ans

$$i) > P(X = -1) = K = 0.1$$

$$ii) > P(X \leq 2) = 0.1 + 0.1 + 0.2 + \cancel{0.2} + 0.3 = 0.9 \quad \text{Ans}$$

$$iii) > P(-1 < X < 3) = \cancel{0.1} + 0.2 + 2 \times 0.1 + 0.3 + \cancel{0.1} = 0.7 \quad \text{Ans}$$

$$iv) > E(X) = -2 \times 0.1 + -1 \times 0.1 + 0 \times 0.2 + 1 \times 2 \times 0.1 + 2 \times 0.3 + 3 \times 0.1 = 0.8 \quad \text{Ans}$$

2022-Fall

3. a) A random variable X has the following probability function,

$X: 2, 3, 4, 5, 6, 7$

$P(X): 0.1, K, 0.2, 2K, 0.3, K$

Find the value of K and find mean and variance.

Here,

$X: 2, 3, 4, 5, 6, 7$

$P(X): 0.1, K, 0.2, 2K, 0.3, K$

Now, By the definition of pmf,

$$\sum_{x} P(x) = 1$$

$$\Rightarrow 0.1 + K + 0.2 + 2K + 0.3 + K = 1$$

$$\Rightarrow 4K + 0.6 = 1$$

$$\Rightarrow 4K = 1 - 0.6 = 0.4$$

$$\therefore K = 0.1$$

Hence, the value of K is 0.1 .

Now,

$$\text{Mean} = E(X)$$

$$= 2 \times 0.1 + 3 \times 0.1 + 4 \times 0.2 + 5 \times 2 \times 0.1 + 6 \times 0.3 + 7 \times 0.1$$

$$= 4.8 \text{ Ans}$$

$$V(X) = E(X^2) - [E(X)]^2$$

where,

$$E(X^2) = 2^2 \times 0.1 + 3^2 \times 0.1 + 4^2 \times 0.2 + 5^2 \times 2 \times 0.1 + 6^2 \times 0.3 + 7^2 \times 0.1$$

$$= 25.2$$

$$\therefore V(X) = (25.2) - (4.8)^2 = 2.16 \text{ Ans}$$

2021-Fall

3.a) A random variable X had the following probability function.

X	-2	-1	0	1	2
$P(X)$	0.2	K	0.4	$2K$	K

i) Find the value of K .

ii) $E(X)$, $E(4X+5)$

iii) $V(X)$, $V(4X+5)$

Solⁿ Here,

X	-2	-1	0	1	2
$P(X)$	0.2	K	0.4	$2K$	K

i) By the definition of probability, The total probability is:

$$\sum_x P(x) = 1$$

$$\Rightarrow 0.2 + K + 0.4 + 2K + K = 1$$

$$\Rightarrow 4K + 0.6 = 1$$

$$\Rightarrow 4K = 1 - 0.6 = 0.4$$

$$\therefore K = 0.1$$

Hence the value of K is 0.1. Ans

$$\begin{aligned} \text{ii) } E(X) &= -2 \times 0.2 + (-1) \times 0.1 + 0 \times 0.4 + 1 \times 2 \times 0.1 + 2 \times 0.1 \\ &= -0.1 \quad \text{Ans} \end{aligned}$$

$$E(4X+5) = 4E(X) + E(5) = 4 \times -0.1 + 5 = 4.6 \quad \text{Ans}$$

$$\text{iii) } V(X) = E(X^2) - [E(X)]^2$$

where,

$$E(X^2) = (-2)^2 \times 0.2 + (-1)^2 \times 0.1 + 0^2 \times 0.4 + 1^2 \times 2 \times 0.1 + 2^2 \times 0.1$$

$$= 1.5$$

$$\therefore V(X) = 1.5 - (-0.1)^2 = 1.49 \text{ Ans}$$

$$\text{Also } V(4X+5) = 4^2 V(X) + V(5)$$

$$= 4^2 \times 1.49 + 0 \quad (\because V(c) = 0)$$

$$= 23.84 \text{ Ans}$$

2019-Spring'

python, NAST

3.a) The following table presents that a discrete probability distribution associated with the daily demand for a product.

No. of demand per day (x) :	10	20	30	40	50	Total
P(x)	0.08	0.24	0.28	0.3	0.1	1

i) Determine the mean daily demand

ii) What is the standard deviation of daily demand?

Here,

x	10	20	30	40	50	Total
P(x)	0.08	0.24	0.28	0.3	0.1	1

i) Mean daily demand:

$$E(x) = 10 \times 0.08 + 20 \times 0.24 + 30 \times 0.28 + 40 \times 0.3 + 50 \times 0.1 \\ = 31 \text{ Ans}$$

ii) The standard deviation of daily demand:

$$S = \sqrt{V(x)} = \sqrt{E(x^2) - [E(x)]^2}$$

where,

$$E(x^2) = 10^2 \times 0.08 + 20^2 \times 0.24 + 30^2 \times 0.28 + 40^2 \times 0.3 + 50^2 \times 0.1 \\ = 1086$$

$$\therefore S = \sqrt{1086 - (31)^2} = 11.18 \text{ Ans}$$

2013-Spring

exb) A coin is tossed until a head appears. What is the expectation of number of tosses required?

Let X be a random variable that denotes the sum of number of tosses required.

$$E(X) = \sum_{i=1}^{\infty} x_i P(x_i)$$

outcomes	sum of number	probability
H	1	$\frac{1}{2}$
HT	2	$\frac{1}{4}$
HHT	3	$\frac{1}{8}$
HHHT	4	$\frac{1}{16}$
$\vdots$	$\vdots$	$\vdots$

Now,

$$E(X) = 1 \times \frac{1}{2} + 2 \times \frac{1}{4} + 3 \times \frac{1}{8} + 4 \times \frac{1}{16} + \dots$$

$$\therefore E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots$$

$$\text{Let } S = E(X) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \quad \text{--- (1)}$$

$$\therefore \frac{S}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots \quad \text{--- (2)}$$

From (1) & (2), we get,

$$S - \frac{S}{2} = \left(\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \right) - \left(\frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots \right)$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4} \right) + \left(\frac{3}{8} - \frac{2}{8} \right) + \left(\frac{4}{16} - \frac{3}{16} \right) + \dots$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} \quad \left(\because S_{\infty} = \frac{t_1}{1-r} \right)$$

$$\Rightarrow \frac{S}{2} = \frac{1}{2} \times 2$$

$$\Rightarrow \frac{S}{2} = 1$$

$$\therefore S = 2$$

Hence, the expected number of ~~req~~ tosses required is 2. Any